Complex systems exist to generate capability. The disciplines around the problem each touch part of it — LENS engineers the interface where they meet, with the human at the center of design and implementation.

A casebook for the Learning Design and Technology program and the Learning Engineering for Next-Generation Systems concentration at the Johns Hopkins University School of Education. A growing record of real incidents — from the bridge of a U.S. Navy destroyer to a Michigan ICU to a national A-Level results algorithm — examined through the lens of capability as a system parameter.

Each case is paired with the learning-engineering insight it carries and the LENS curriculum it informs. Together they form an evidence base for the argument that capability engineering is a discipline, not an afterthought.

Learning engineering is a **team sport** — a dialogue between engineering and the learning sciences, open to every field that brings a real tool. Bring what you know: domain, code, theory, design, teaching, analysis. No one knows everything. You get to decide what you bring.

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A CASEBOOK

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JOHNS HOPKINS UNIVERSITY

LEARNING DESIGN AND TECHNOLOGY